



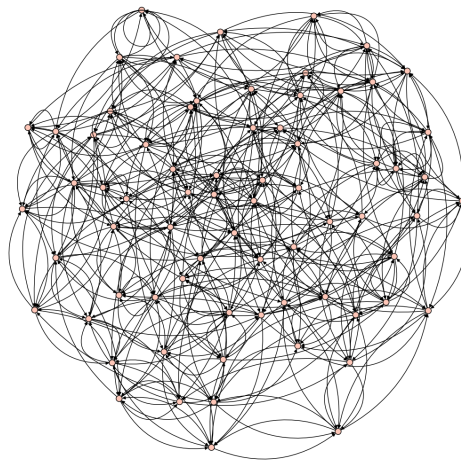
THE UNIVERSITY OF
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FINAL REPORT

The Art of Scientific Computation: Spectral and Mixing Properties of Supersingular Isogeny Graphs

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Chapter 1

Introduction

1.1 Overview

In this project, I will construct and analyse supersingular ℓ -isogeny graphs over finite fields $\mathbb{F}_{p^2}$. These are graphs whose vertices represent isomorphism classes of supersingular elliptic curves over $\mathbb{F}_{p^2}$ and whose edges represent degree- ℓ isogenies between them (see [5] for background). These graphs are of special interest in cryptography, since many cryptographic protocols can be understood as applying random walks on these graphs. One such example is given in [1], whose designs provide inspiration for the graphs considered in this project.

The significance of these graphs in cryptography arises mainly from their strong expansion properties; in particular, they are Ramanujan graphs, and therefore random walks on these graphs converge quickly to a uniform distribution. The most cryptographically interesting consequence is that given any two vertices in one such graph, it will be computationally expensive to find a path connecting them.

We are therefore interested in performing empirical verification of theoretical properties attributed to these graphs. To this end, two main interests stand out: the eigenvalue spectra associated to the adjacency matrices of these graphs, and the mixing times of random walks on these graphs. For this project, I will produce a sample of supersingular isogeny graphs over a range of parameters and compute these two characteristics, comparing them to theoretical estimates in order to illustrate and corroborate them.

1.2 Scope

We have the following primary objectives.

1. Efficiently compute supersingular ℓ -isogeny graphs over $\mathbb{F}_{p^2}$ for a range of parameters p and ℓ .
2. Compute the adjacency matrices for a sample of supersingular isogeny graphs and compute their eigenvalue spectra. We will use this to answer two questions.
 - (a) How does the spectral gap of a supersingular ℓ -isogeny graph over $\mathbb{F}_{p^2}$ vary according to the parameters ℓ and p ?
 - (b) How does the spectral gap of a supersingular ℓ -isogeny graph over $\mathbb{F}_{p^2}$, as measured empirically, align with theoretical estimates of the same quantity?
3. Produce, for each sampled supersingular ℓ -isogeny graph over $\mathbb{F}_{p^2}$, a sample of random walks on that graph. We will use this to answer two questions.
 - (a) How are mixing times of random walks distributed for each graph, in terms of parameters ℓ and p ?
 - (b) How do the mixing times of random walks on these graphs, as measured empirically, align with theoretical estimates of mixing times for Ramanujan graphs?

Chapter 2

Mathematical Background

This chapter will cover some of the relevant concepts and vocabulary for this project, as well as an outline of the properties we will wish to assess. While some background in group theory is assumed*, we aim to remain light on technicalities, focusing mainly on the key terms required to understand the problem at hand. The interested reader may consult [5], [2], or [4] for further information; indeed, the following can be treated as an abbreviated summary of these sources. The sections on spectral graph theory, and on random walks and mixing times, are drawn from [3].

2.1 Elliptic curves

The definition of elliptic curve can be rendered in various equivalent forms. For instance, as per [6], we could define an elliptic curve E over a field K as a non-singular projective algebraic curve of genus 1 having a distinguished K -rational point $O \in E$. But, we are interested in elliptic curves with a view to computation, and therefore give a simpler definition, more tractable for our purposes.

Definition 1 (Elliptic Curve). Let K be a field of characteristic other than 2 or 3. Then an elliptic curve E over K consists of the set of solutions $(x, y) \in K^2$ of the short Weierstrass equation

$$E : y^2 = x^3 + ax + b$$

where $a, b \in K$ and $4a^3 + 27b^2 \neq 0$.

Over the real numbers $\mathbb{R}$, elliptic curves can be viewed as familiar planar curves.

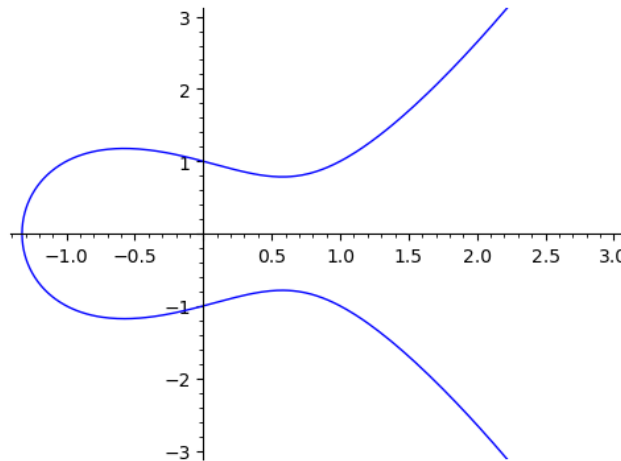


Figure 2.1: Elliptic curve $E : y^2 = x^3 - x + 1$ over $\mathbb{R}$.

However, we will be mainly interested in elliptic curves over finite fields $\mathbb{F}_{p^n}$ for prime p and $n \in \mathbb{Z}_{\geq 1}$. An example of the solution set to one such curve over a finite field is shown below.

*If the reader is interested in a review of the relevant definitions, sources such as [7] will provide a succinct summary.

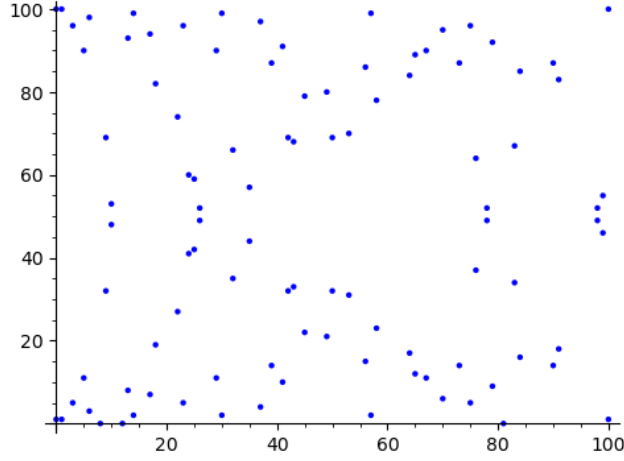


Figure 2.2: Elliptic curve $E : y^2 = x^3 - x + 1$ over $\mathbb{F}_{101}$, the finite field of order 101.

Key to the theory of elliptic curves is the fact that it is possible to define a group operation (conventionally called a *group law*) on the points of E , satisfying certain desirable properties. The definition of this operation arises from projective geometry, whose details we defer to a reference such as [6]. For our purposes, it will suffice to understand that any elliptic curve has a group structure.

Definition 2 (Group Law). Let E be an elliptic curve over a field K . It is a consequence of Bézout’s theorem that under mild assumptions and counting multiplicity, an algebraic curve of degree m and an algebraic curve of degree n will intersect in mn points. Since an elliptic curve has degree 3 and a line has degree 1, it follows that the line spanned by two points $P, Q \in E$ will intersect E at a unique third point $R' \in E$. We let $R \in E$ be the reflection of R' over the x -axis and define a *group law* $P \oplus Q = R$, such that $(E, \oplus)$ forms an abelian group. (For details, see [6].)

2.2 Isogenies

If E_1 and E_2 are elliptic curves over a common field K , then a morphism of elliptic curves is a rational map $f : E_1 \rightarrow E_2$. Loosely, this can be understood as a map defined by rational functions. However, we will be interested in a more restrictive notion of mapping between elliptic curves, namely *isogeny*.

Definition 3 (Isogeny). Let E_1 and E_2 be elliptic curves over a common field K . An *isogeny* of elliptic curves is a morphism $f : E_1 \rightarrow E_2$ which is also a surjective group homomorphism with respect to the group structures on E_1 and E_2 .

Isogenies provide a notion of mapping between elliptic curves which respects both the geometric and arithmetic structures of the curve, and which can therefore encode useful information about the curves thereby related. We will enumerate some useful properties.

Definition 4 (Elliptic curve isomorphism). An isogeny $f : E_1 \rightarrow E_2$ is called an *isomorphism* if it is an isomorphism of the underlying groups in E_1 and E_2 .

Definition 5 (Degree of an isogeny). Suppose an isogeny $f : E_1 \rightarrow E_2$ is separable[†]. Then f has a quantity called the *degree*, denoted by $\deg(f)$, which can be computed by $\deg(f) = |\ker(f)|$.

In fact, over suitable fields, the kernel of an isogeny will uniquely determine that isogeny.

We next consider the possibility of defining an isogeny between a curve and itself.

Definition 6 (The endomorphism ring). Let E be an elliptic curve over a field K . An isogeny $f : E \rightarrow E$ is called an endomorphism of E . The set $\text{End}(E)$ of all endomorphisms of E , including also the zero morphism $0 : E \rightarrow E$, forms a ring where addition is defined pointwise and multiplication is defined to be the composition operation. That is, $fg = f \circ g$ and for $f, g \in \text{End}(E)$ we define $f + g$ to be the isogeny $x \mapsto f(x) + g(x)$. The resulting ring $\text{End}(E)$ is called the *endomorphism ring* of E .

[†]For our purposes, all isogenies will be separable. The reader can therefore, if so desired, treat this condition as a technicality.

Since any elliptic curve E forms an abelian group, we can define integer multiplication of points on the curve using the additive group structure. Then, it holds for any elliptic curve E that the multiplication-by- n map $x \mapsto nx$ is an endomorphism $[n]: E \rightarrow E$. So, for any elliptic curve E , $\text{End}(E)$ has a subring isomorphic to $\mathbb{Z}$.

2.3 Ordinary and supersingular

Elliptic curves over finite fields can be distinguished as either *ordinary* or *supersingular*. Ordinary elliptic curves form well understood structures; we will be interested in the alternate, supersingular case.

Definition 7. Let E be an elliptic curve over a finite field K . If $\text{End}(E)$ is isomorphic to an order in an imaginary quadratic field $\mathbb{Q}(\sqrt{-d})$, we say that E is *ordinary*. In this case, $\text{End}(E)$ is commutative.

If instead $\text{End}(E)$ is isomorphic to an order in a quaternion algebra, we say that E is *supersingular*. In this case, $\text{End}(E)$ is non-commutative.

All elliptic curves over finite fields are either ordinary or supersingular, and we see that supersingular curves can be distinguished by having larger endomorphism rings. This can be understood, morally, as indicating that supersingular elliptic curves have more complex internal structure.

Property 1. Over any field of characteristic p , if E is a supersingular elliptic curve then the j -invariant $j(E)$ is defined over $\mathbb{F}_{p^2}$. And, there are $N_p = \lfloor \frac{p}{12} \rfloor + \varepsilon_p$ such supersingular elliptic curves over characteristic p , where $\varepsilon_p = 0, 1, 1, 2$ corresponding to the cases that $p \equiv 1, 5, 7, 11 \pmod{12}$ respectively.

Supersingular elliptic curves are named so because they are singular in the sense of rarity; over any field, few supersingular elliptic curves exist compared to ordinary elliptic curves.

2.4 The j -invariant

Isomorphisms are typically regarded as providing a notion of *equivalence*; if two objects are isomorphic within some category, then their properties within that category must be identical. So, we are often interested in classifying objects *up to isomorphism*. For elliptic curves, isomorphism classes of elliptic curves have a convenient classification by the j -invariant. This invariant has a more general definition, but we give a specific form applicable to curves expressed by Weierstrass equations.

Definition 8 (j -invariant). Let E be an elliptic curve over a field K , and assume that E is given by the Weierstrass equation $y^2 = x^3 + ax + b$. Then the j -invariant of E is

$$j(E) = 1728 \frac{4a^3}{4a^3 + 27b^2}.$$

When the base field K is algebraically closed, such as in the case that $K = \mathbb{C}$, then two elliptic curves E_1, E_2 over K are isomorphic if and only if $j(E_1) = j(E_2)$. So, in this case, the j -invariant classifies all elliptic curves up to isomorphism.

Later we will consider elliptic curves over $\mathbb{F}_{p^2}$, where p is prime. In this case, although $\mathbb{F}_{p^2}$ is not algebraically closed, every supersingular elliptic curve has its j -invariant defined as an element of $\mathbb{F}_{p^2}$, and therefore the j -invariant can still be used as a classification of isomorphism classes in the algebraic closure of $\mathbb{F}_{p^2}$.

2.5 Supersingular ℓ -isogeny graphs

An *isogeny graph* is, in general, a graph whose vertices correspond to isomorphism classes of elliptic curves, and whose edges are isogenies between those (isomorphism classes of) elliptic curves. We are interested in certain *supersingular* isogeny graphs, whose edges are restricted to isogenies of common fixed degree.

Definition 9 (ℓ -isogeny graphs). Let K be a field. For $\ell \in \mathbb{Z}_{\geq 0}$, the ℓ -isogeny graph over K is the graph $G_\ell(K)$ whose vertices are isomorphism classes of elliptic curves over K , and whose edges are ℓ -isogenies (i.e. isogenies of degree ℓ).

We will hereafter restrict our attention to the case that $K = \mathbb{F}_{p^2}$, where $p \in \mathbb{Z}$ is prime. In this case, the supersingular elliptic curves over $\mathbb{F}_{p^2}$ form a connected component within $G_\ell(\mathbb{F}_{p^2})$, having approximately $p/12$ vertices. As we will see, this graph has many desirable properties.

Definition 10 (Supersingular ℓ -isogeny graphs). Let $\ell, p \in \mathbb{Z}$ be primes such that $\ell \neq p$. The set of supersingular elliptic curves over $\mathbb{F}_{p^2}$ forms a connected component within $G_\ell(\mathbb{F}_{p^2})$, and is called the *supersingular ℓ -isogeny graph over $\mathbb{F}_{p^2}$* , or alternatively the *supersingular component of $G_\ell(\mathbb{F}_{p^2})$* .

Property 2. *Any supersingular ℓ -isogeny graph over $\mathbb{F}_{p^2}$ is finite, connected and $(\ell + 1)$ -regular.*

2.6 Spectral graph theory

Spectral graph theory pursues the insight that the various matrices associated with a graph, such as the adjacency matrix, encode in their eigenvalue spectra useful information pertaining to their underlying graphs.

Definition 11 (Adjacency matrix). Let G be a graph with vertex set $V = \{v_1, \dots, v_n\}$ and edge set E . The adjacency matrix of G is the $n \times n$ matrix A whose (i, j) -th entry a_{ij} is equal to the number of edges in E beginning at v_i and ending at v_j .

It is a simple result of linear algebra that if the adjacency matrix A of a graph is symmetric, then its eigenvalues are all real. In particular, this will hold for either undirected graphs, or for symmetric directed graphs.

Definition 12 (Ramanujan graph). Suppose G is a connected, k -regular graph with n vertices and adjacency matrix A . Suppose the eigenvalues $\lambda_1, \dots, \lambda_n$ are sorted by decreasing absolute value,

$$|\lambda_1| \geq |\lambda_2| \geq \dots \geq |\lambda_{n-1}| \geq |\lambda_n|.$$

Then G is a *Ramanujan graph* if

$$|\lambda_2| \leq 2\sqrt{k-1}.$$

In particular, in the case that G has symmetric adjacency matrix A , the eigenvalues of A satisfy

$$k = \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_{n-1} \geq \lambda_n \geq -k.$$

Now, recalling that the supersingular ℓ -isogeny graph over $\mathbb{F}_{p^2}$ is connected and $(\ell + 1)$ -regular, we have the following property.

Property 3. *The supersingular ℓ -isogeny graph over $\mathbb{F}_{p^2}$ is a Ramanujan graph, satisfying*

$$|\lambda_2| \leq 2\sqrt{\ell}.$$

2.7 Random walks and mixing times

We have seen that supersingular ℓ -isogeny graphs over $\mathbb{F}_{p^2}$ are Ramanujan graphs. The most important consequence of this property is that random walks on these graphs have optimal mixing times.

Definition 13 (Random walk). Let G be a graph. Supposing a vertex v of G has at least one outgoing edge, a *random step* on this graph starting v is a uniformly random choice from the set of neighbours of v (possibly including v itself, in the case that G contains self-loops). A *random walk* on G is a finite or countable sequence of random steps.

It follows from the fact that supersingular ℓ -isogeny graphs are Ramanujan graphs that they satisfy the following essential property (see [3, Theorem 1] for proof).

Property 4 (Random walk theorem). *Let j_0 be the j -invariant corresponding to a supersingular elliptic curve in $\mathbb{F}_{p^2}$ and let j be the j -invariant corresponding to the final point reached after a random walk of n steps in the supersingular ℓ -isogeny graph. Then for every $j' \in \mathbb{F}_{p^2}$ we have*

$$\left| \Pr(j = j') - \frac{1}{N_p} \right| \leq \left(\frac{2\sqrt{\ell}}{\ell + 1} \right)^n.$$

Indeed, the stationary distribution for random walks on any k -regular graph converges to the uniform distribution, and the above theorem guarantees that for supersingular ℓ -isogeny graphs, the rate of this convergence is exponential, increasing with greater ℓ .

In light of this, let us consider a random walk on a supersingular ℓ -isogeny graph beginning at j -invariant j_0 . Defining $P_k \in \mathbb{R}^{N_p}$ to be the row vector with a 1 in the k -th entry and zeroes in every other entry, and defining

$$P_n(i) = \Pr(\text{walk is at } j_i \text{ after } n \text{ steps}),$$

we find that the vectors P_n for $n \geq 0$ describe the probability distributions for the destinations of random walks on this graph after n steps.

Now, it is a standard result of the theory of discrete-time Markov chains that our random walk probability distribution evolves according to the rule $P_{n+1} = P_n M$ where $M = \frac{1}{1+\ell}A$, given that A is the adjacency matrix of our supersingular ℓ -isogeny graph. In particular, it follows that $P_n = P_0 M^n$.

Our claim is that P_n converges to the following uniform distribution in a suitable sense.

Definition 14 (Uniform distribution). Recall that a supersingular ℓ -isogeny graph has N_p vertices. So, for any j -invariant j_i in this graph, the uniform distribution assigns to j_i the probability $U(i) = \frac{1}{N_p}$.

Our preferred measure of convergence will be the following.

Definition 15 (Total variation distance). The *total variation distance* between the probability distributions P_n and U is defined by

$$d_{TV}(P_n, U) = \frac{1}{2} \sum_{i=1}^{N_p} \left| P_n(i) - \frac{1}{N_p} \right|.$$

This can be quantified by the following *mixing time*, which we will later numerically analyse.

Definition 16 (Mixing time). For $\varepsilon > 0$, the *mixing time* $t_{\text{mix}}(\varepsilon)$ quantifies convergence of P_n to U using

$$t_{\text{mix}}(\varepsilon) = \min\{n \geq 0 \mid d_{TV}(P_n, U) \leq \varepsilon\}.$$

Chapter 3

Computation

3.1 Elliptic curves in Sage

Sage is a software package built atop Python which is widely used in mathematical research for algebraic computation. Its facilities include:

- representing finite fields as Python objects;
- representing elliptic curves as Python objects;
- testing elliptic curves for supersingularity;
- computing isogenies between elliptic curves;
- computing j -invariants corresponding to elliptic curves;
- graph structures;

and more, which will be relevant to our use case. In our project, we will use Sage to handle heavier algebraic operations so that we can focus on the relevant computational research questions.

3.2 Generating supersingular ℓ -isogeny graphs

In constructing supersingular ℓ -isogeny graphs, we would like vertices to represent isomorphism classes of supersingular elliptic curves over $\mathbb{F}_{p^2}$. But, we have seen that (up to certain technical details), each such isomorphism class is uniquely specified by a j -invariant, taking a value in $\mathbb{F}_{p^2}$. So, rather than using any concrete representation of an elliptic curve, we can perform all our required elliptic curve operations in terms of values in $\mathbb{F}_{p^2}$, namely j -invariants. Indeed, given a value `j_val` in a Sage finite field $\mathbb{F}_{p^2}$, one obtains a Boolean

```
EllipticCurve(j=j_val).is_supersingular()
```

corresponding to whether a supersingular elliptic curve over $\mathbb{F}_{p^2}$ exists having j -invariant equal to `j_val`. So, j -invariants will provide vertex representations for our isogeny graphs.

Now, since the supersingular ℓ -isogeny graph is known to be connected, it will suffice to first identify one supersingular j -invariant over $\mathbb{F}_{p^2}$, and then to perform a graph traversal to compute the rest of the graph. In our implementation, we have chosen to use a breadth-first search.

It still remains to understand, however, how we will compute the isogeny-neighbours of any such j -invariant. That is, given a j -invariant j_0 known to lie in our graph, how will we find the set of j -invariants j_i for $i = 1, \dots, \ell + 1$ such that there exists an isogeny $j_0 \rightarrow j_i$? There are two main options before us.

First, recall that in suitable (‘separable’) cases it is possible uniquely determine an isogeny by its kernel. And, since any isogeny $f: E_1 \rightarrow E_2$ is a group homomorphism, this kernel is a subgroup of E_1 ; it follows that an isogeny can be identified by finding the generators of its kernel. Supposing this can be done, one may use *Vélu’s formulas* to compute explicitly the elliptic curve E_2 which is the image

of an isogeny f having the specified kernel. And, indeed, the facilities for this are provided in Sage. But, this will typically be more computationally expensive than we desire, especially since an initial sampling process is required in order to find generators for suitable subgroups in E_1 .

So, we will instead use an algebraic device known as the *modular polynomial*. This is an integer valued polynomial in two variables $\Phi_\ell \in \mathbb{Z}[X, Y]$ such that (X, Y) is a root of Φ if and only if there exists an ℓ -isogeny between elliptic curves $E_1 \rightarrow E_2$ such that $j(E_1) = X$ and $j(E_2) = Y$. In the case that we consider j -invariants over finite fields such as $\mathbb{F}_{p^2}$, the same result is obtained by evaluating $\Phi_\ell(X, Y)$ over that field instead. So, suppose we know at least one j -invariant, say j_0 , lying in our graph. In order to compute its ℓ -isogeny neighbours then, it will suffice to find the roots of the polynomial $\Phi_\ell(j_0, Y)$. Indeed, in sage one can construct the modular polynomial by writing the following.

```
Phi = classical_modular_polynomial(Integer(1))
```

Then, for any known j -invariant j_1 lying in our graph, we can construct a polynomial

```
poly_y = Phi(j1, Y)
```

and find the neighbouring j -invariants of j_1 by computing the roots of `poly_y`, enumerating as follows.

```
for j2, mult in poly_y.roots():
    . . .
```

This will be our instrument for finding the neighbours of any vertex in our isogeny graph.

Now that we have established representations for the vertices in our graph, as well as for computing the neighbours of any vertex in our graph, there remain two computational steps. The first is to identify at least one initial supersingular j -invariant over $\mathbb{F}_{p^2}$, and the second is to perform a graph traversal beginning at that initial j -invariant. But, since $\mathbb{F}_{p^2}$ contains precisely p^2 elements, for moderate p it will be possible simply to enumerate the elements of $\mathbb{F}_{p^2}$, and check each element for supersingularity as a j -invariant. Once an example is found, we can perform a breadth-first search using the modular polynomial to find neighbouring vertices in order to produce the entire supersingular ℓ -isogeny graph.

3.2.1 An aside on exceptional j -invariants

We earlier claimed that for any prime p and any prime $\ell \neq p$, the supersingular ℓ -isogeny graph is $(\ell + 1)$ -regular. This is true with a minor caveat: for some primes p , the elliptic curves having j -invariants 0 or 1728 may be supersingular, and may therefore be elements of our graph. But, these two (isomorphism classes of) elliptic curves are known to be exceptional, being the only two elliptic curves possessing what are known as ‘extra automorphisms’. Consequently, if these are elements in our graph, the graph will fail to be $(\ell + 1)$ -regular at these two points. Computationally this is minimally consequential, but for simplicity we will prefer to avoid this case altogether.

Now, it is known that the j -invariant 0 is supersingular over $\mathbb{F}_{p^2}$ if and only if $p \equiv 2 \pmod{3}$, and that the j -invariant 1728 is supersingular over $\mathbb{F}_{p^2}$ if and only if $p \equiv 3 \pmod{4}$. So, to avoid these cases, we will perform our experiments specialising to the cases that p is a prime with $p \equiv 1 \pmod{12}$. To do this, we will write a function `first_n_suitable_primes` which returns a list of the first n primes which are congruent to 1 modulo 12, beginning with 13. This function will be computed by simple enumeration, using primality checking provided by Sage.

3.3 Graph analysis

Once the supersingular ℓ -isogeny graph has been constructed (hereafter denoted $G_\ell^s(\mathbb{F}_{p^2})$ for short), the remainder of the project focuses on extracting useful quantitative information from this graph. The graph $G_\ell^s(\mathbb{F}_{p^2})$ has a combinatorial aspect, encoded by its adjacency matrix A , as well as a stochastic aspect given by its random walks. These two aspects are closely related, and we will like to examine this relationship quantitatively.

The adjacency matrix A can be evaluated using standard tools of linear algebra. Since A has size $N_p \times N_p$, and we saw in Property 1 that $N_p \approx p/12$, we expect that this will be computationally feasible

for p up to the range of 10^3 to 10^4 . And indeed, we see using our earlier `first_n_suitable_primes` function that the 200-th prime which is congruent to 1 modulo 12 is 6529, and that the 300-th such prime is 9973.

Then, we will examine the mixing properties of $G_\ell^s(\mathbb{F}_{p^2})$ by two means. First, we can use the relation $P_n = P_0 M^n$ earlier ascertained for $M = \frac{1}{\ell+1}A$ in order to compute the probability distribution for random walks of length n at once. This can be done using standard computational linear algebra facilities. By these means, we will compute the mixing times $t_{\text{mix}}(\varepsilon)$ for a range of ε .

To make this concrete, however, we will also compute a sample of random walks beginning at a common initial vertex and plot the distribution of their endpoints. This will be done by using standard graph random walk procedures.

Chapter 4

Results

Remark. I would like to pre-emptively apologise for the poor spacing of figures in what follows.

We have generated in total nine hundred supersingular ℓ -isogeny graphs $G_\ell^s(\mathbb{F}_{p^2})$. These cover three isogeny degrees, $\ell = 2, 3, 5$, and three hundred primes p , namely each of the first 300 primes which are congruent to 1 modulo 12. These 300 primes range from 13 to 9973, and therefore present a sample across several order of magnitude; this was chosen so as to get a sense for any scaling-related variation in observations.

All resultant data can be found in CSV files, stored under the directory `/data/results/` in the codebase. The following relevant experiments were conducted.

- General graph structure: We evaluated the basic properties of each $G_\ell^s(\mathbb{F}_{p^2})$ and verified that they satisfied the expected elementary graph theoretic properties, for instance $(\ell + 1)$ -regularity and that the number of vertices N_p grows approximately as $p/12$. The relevant data can be found in `spectral_results.csv`.
- Spectral analysis: We conducted spectral analysis, verifying that the largest eigenvalue of each adjacency matrix A is exactly $\ell + 1$, then comparing this value with the second largest eigenvalue of A , plotting the resultant *spectral gap* against p . We verified that in all cases, the Ramanujan bound $|\lambda_2| \leq 2\sqrt{\ell}$ was satisfied. Convergence toward the Ramanujan bounds was plotted over varying p and ℓ . The relevant data can be found in `spectral_results.csv`.
- Exact mixing times: Since it was established that the distribution of random walks of n steps can be computed precisely using the relation $P_n = P_0 M^n$ where $M = \frac{1}{\ell+1}A$, with A being the adjacency matrix, of the relevant graph, we were able to compute the precise total variation distances over time, and thereby mixing times, for varying p and ℓ . The results were plotted against both p and $\log(N_p)$, where the mixing times $t_{\text{mix}}(\varepsilon)$ were evaluated for $\varepsilon \in \{0.1, 0.05, 0.01\}$. The relevant data can be found in `tv_curve_results.csv` and `mixing_results.csv`.
- Random walks and empirical total variation distance: In order to obtain a better sense for the exact mixing time results computed using the distribution P_n , we also simulated for each $\ell \in \{2, 3, 5\}$ and $p \in \{4789, 9973\}$ a sample of 5000 random walks for lengths

$$n \in \{1, 2, 3, 5, 10, 20, 30, 50\}.$$

This was done in order to observe empirical total variation distance as the lengths of walks increased. An additional 5000 walks of length 50 were sampled for each of the above graphs in order to produce a histogram of endpoints after 50 steps. This is to say that for each of the six graphs $G_\ell^s(\mathbb{F}_{p^2})$ specified, we computed 45,000 random walks of varying lengths. The relevant data can be found in `tv_curve_results.csv`, `empirical_random_walk_results.csv` and `endpoint_histogram_results.csv`.

Remark. In fact, the $p = 13$ case was also computed, pending the later realisation that the results would be totally uninformative; only one supersingular vertex exists over $p = 13$. Thankfully, for the same reason that the resultant data was uninformative, it was also computationally negligible. We will simply omit its discussion.

In total, approximately 60 minutes of computation on an M1 Macbook Air was required in order to generate all graphs and complete all experiments. This duration includes, besides producing all the CSV files and plots included in the codebase, a selection of 275 graphical renders of supersingular ℓ -isogeny graphs. These total over 100MB and so have not been included in my submission, but are available upon request. A small selection will be shown shortly.

4.1 Visual isogeny graphs

Here is a selection of visualisations of the supersingular ℓ -isogeny graphs we have produced, for varying p and ℓ . They have been rendered using Sage's native plotting functionality.

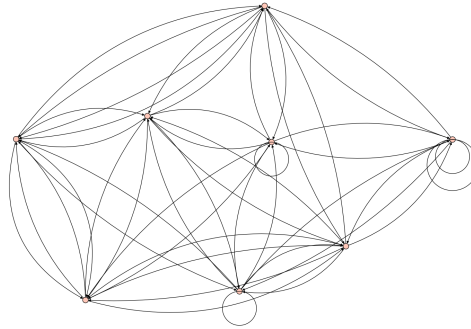


Figure 4.1: Graph $G_{\ell}^s(\mathbb{F}_{p^2})$ where $p = 97$ and $\ell = 5$.

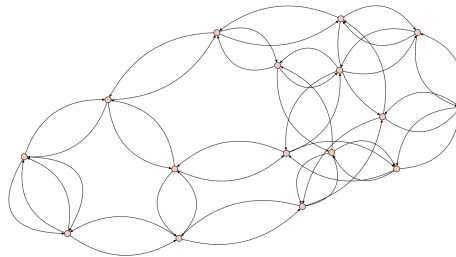


Figure 4.2: Graph $G_{\ell}^s(\mathbb{F}_{p^2})$ where $p = 193$ and $\ell = 2$.

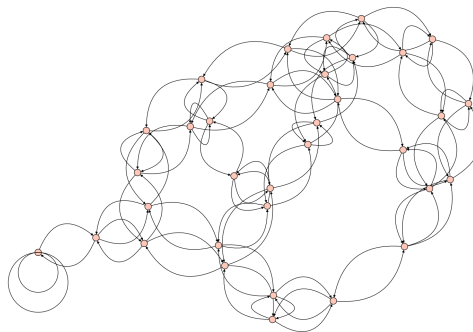


Figure 4.3: Graph $G_{\ell}^s(\mathbb{F}_{p^2})$ where $p = 409$ and $\ell = 2$.

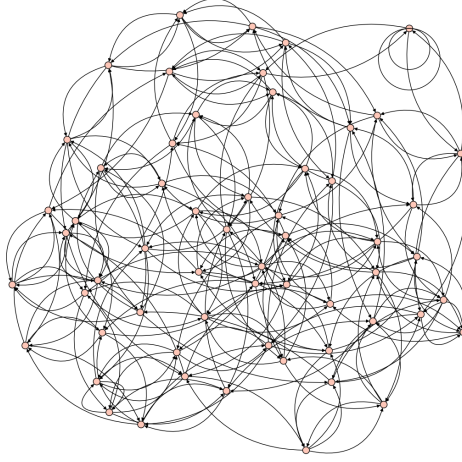


Figure 4.4: Graph $G_\ell^s(\mathbb{F}_{p^2})$ where $p = 709$ and $\ell = 3$.

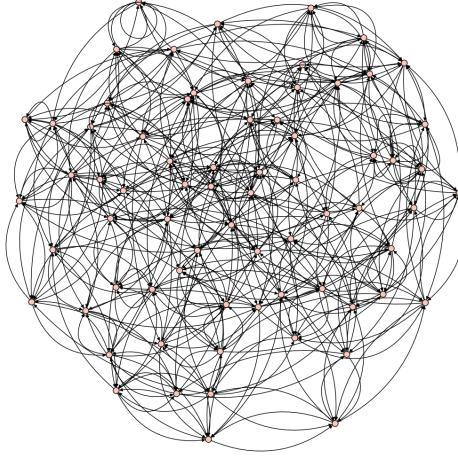


Figure 4.5: Graph $G_\ell^s(\mathbb{F}_{p^2})$ where $p = 853$ and $\ell = 5$.

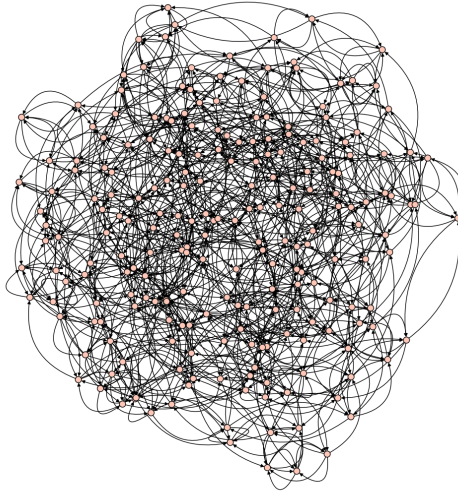


Figure 4.6: Graph $G_\ell^s(\mathbb{F}_{p^2})$ where $p = 2689$ and $\ell = 3$.

4.2 General graph structure

We first observe that Property 1, describing the approximation $N_p \approx p/12$ where N_p is the number of isomorphism classes of supersingular elliptic curves over $\mathbb{F}_{p^2}$, is clearly corroborated.

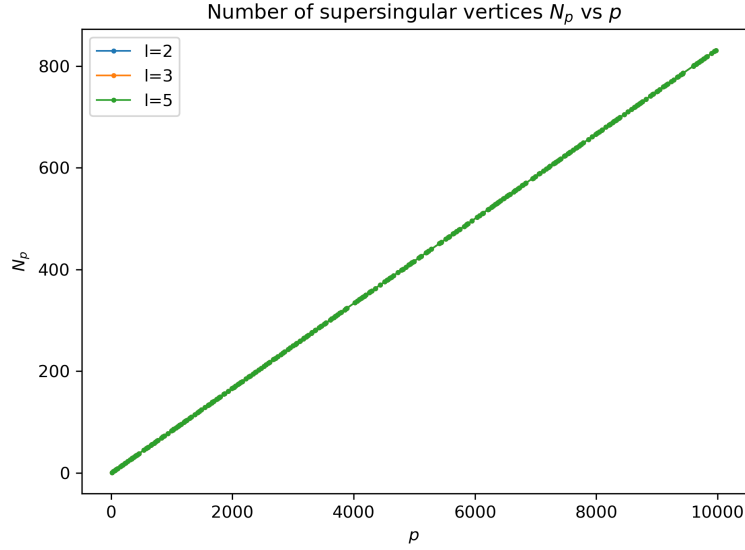


Figure 4.7: N_p plotted against p .

Indeed, the degree ℓ does not affect the value N_p , and this can be seen in Figure 4.7. A very clear linear relationship is visible.

It was verified that all isogeny graphs we produced were $(\ell + 1)$ -regular and connected, but since this is less amenable to visualisation, we will direct the reader to see `spectral_results.csv` for confirmation if desired.

4.3 Spectral analysis

Recall that if $G_\ell^s(\mathbb{F}_{p^2})$ has adjacency matrix A then the eigenvalues of A can be arranged

$$\ell + 1 = |\lambda_1| \geq |\lambda_2| \geq \cdots \geq |\lambda_{n-1}| \geq |\lambda_n| \geq -(\ell + 1).$$

Then, we see in Figure 4.8 that $|\lambda_2|$ approaches the Ramanujan bound $|\lambda_2| \leq 2\sqrt{\ell}$ with increasing p .

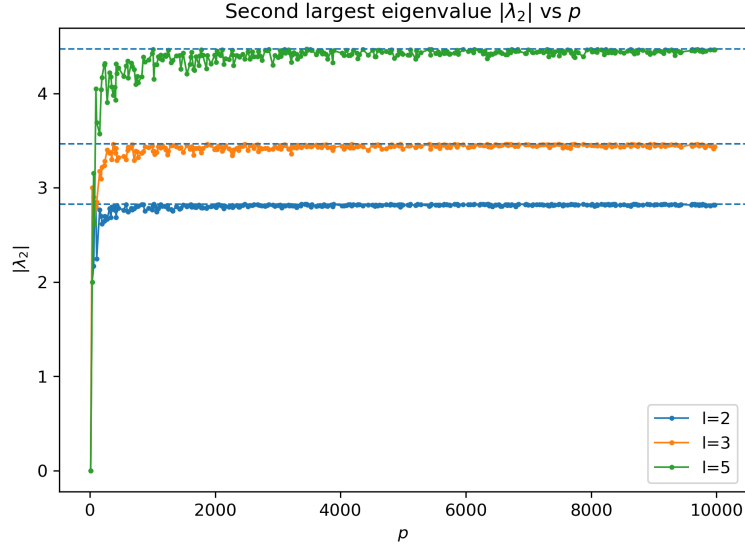


Figure 4.8: Values of $|\lambda_2|$ plotted against p . Here, the three dotted lines correspond to each of the three Ramanujan bounds $2\sqrt{\ell}$ for $\ell = 2, 3, 5$.

Indeed, the Ramanujan bound suggests that there should exist an inequality $|\lambda_2|/2\sqrt{\ell} \leq 1$. To see this, we plot the ‘*Ramanujan ratio*’ $|\lambda_2|/2\sqrt{\ell}$ against p in Figure 4.9.

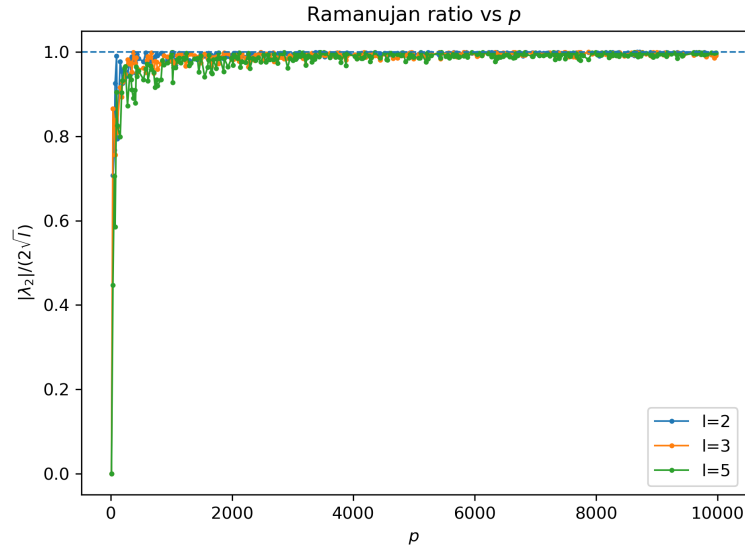


Figure 4.9: Values of the Ramanujan ratio $|\lambda_2|/2\sqrt{\ell}$ plotted against p .

As we see, there exists rapid convergence toward the ratio imposed by the Ramanujan bound. This is interesting because it suggests that not only is the Ramanujan bound true, but that it is a *minimal* bound on $|\lambda_2|$.

We finally observe in Figure 4.10 another quantity which is of great general interest in spectral graph theory, namely the *spectral gap* $|\lambda_1| - |\lambda_2|$. For graphs $G_\ell^s(\mathbb{F}_{p^2})$, this is known to be equal to $(\ell + 1) - |\lambda_2|$.

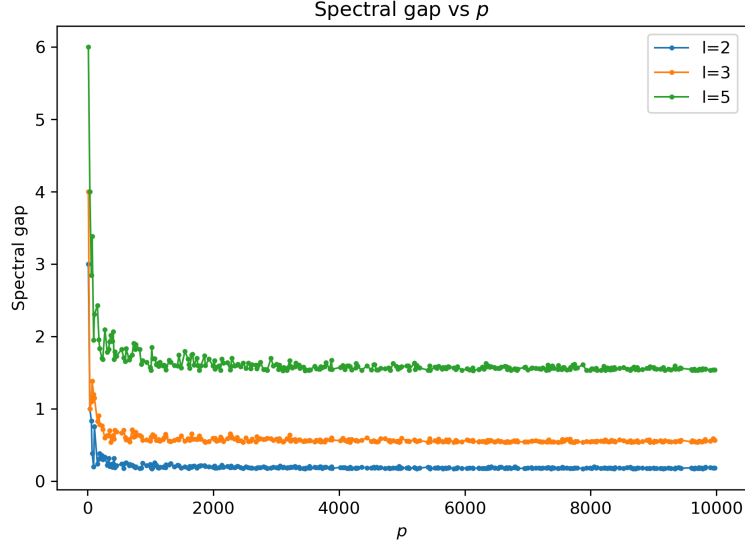


Figure 4.10: Values of the spectral gap $(\ell + 1) - |\lambda_2|$ plotted against p .

Here, we observe some interesting behaviour: while the spectral gap appears to decrease approximately monotonically for all $\ell \in \{2, 3, 5\}$, we witness that larger ℓ seems to correspond with larger spectral gaps. In spectral graph theory, this is known to correspond with the quality of greater connectivity. Certainly, this coheres with the intuition that by increasing the number of edges in a regular graph, one should also increase connectivity.

It is also interesting to make an observation in relation to the earlier plot of the Ramanujan ratio. Namely, our observations of the Ramanujan ratio suggest an asymptotic relation $|\lambda_2| \approx 2\sqrt{\ell}$ as p increases. This is to say that $|\lambda_2|$ has a square-root relation with ℓ . But, since the spectral gap is defined by $(\ell + 1) - |\lambda_2|$, it follows that we expect the relation between the spectral gap and ℓ to follow a curve

$$f(\ell) = \ell - 2\sqrt{\ell} + 1$$

as p increases. Or rather, we might conjecture on the basis of this empirical evidence that if we define the spectral gap $\text{spect}(\ell, p) = \text{spectral gap of } G_\ell^s(\mathbb{F}_{p^2})$, we should expect to see that

$$\lim_{p \rightarrow \infty} \text{spect}(\ell, p) = f(\ell)$$

Here, f is monotonically increasing for $\ell \geq 0$, and we therefore obtain analytic justification for the observation that increasing ℓ will tend to increase the spectral gap. This provides an alternative perspective on the increasing connectivity of $G_\ell^s(\mathbb{F}_{p^2})$ for large ℓ .

4.4 Exact mixing times

Recall that $P_n \in \mathbb{R}^{N_p}$ describes the distribution of probabilities that a random walk beginning at some initial vertex on $G_\ell^s(\mathbb{F}_{p^2})$ will land on each other vertex after n steps. Recall too that we have the analytical relation

$$P_n = P_0 M^n \quad \text{for} \quad M = \frac{1}{\ell + 1} A,$$

with A the adjacency matrix of $G_\ell^s(\mathbb{F}_{p^2})$ and P_0 the vector with 1 in the entry corresponding to the initial vertex and 0 in all other entries. Then, we can numerically quantify the convergence of P_n to the uniform distribution U for increasing n using the total variation distance

$$d_{TV}(P_n, U) = \frac{1}{2} \sum_{i=1}^{N_p} \left| P_n(i) - \frac{1}{N_p} \right|$$

and the mixing time

$$t_{\text{mix}}(\varepsilon) = \min\{n \geq 0 \mid d_{\text{TV}}(P_n, U) \leq \varepsilon\},$$

where $\varepsilon > 0$. In our experiment, we have chosen to study the cases that $\varepsilon \in \{0.1, 0.05, 0.01\}$.

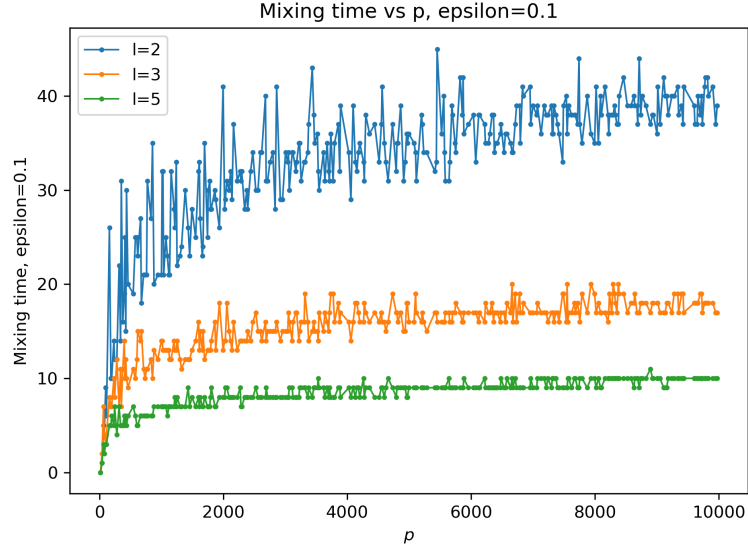


Figure 4.11: Mixing times $t_{\text{mix}}(0.1)$ plotted against increasing p .

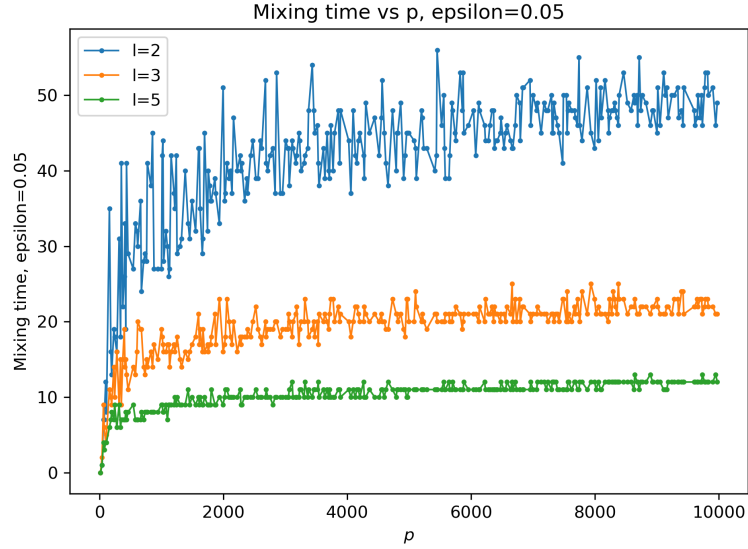


Figure 4.12: Mixing times $t_{\text{mix}}(0.05)$ plotted against increasing p .

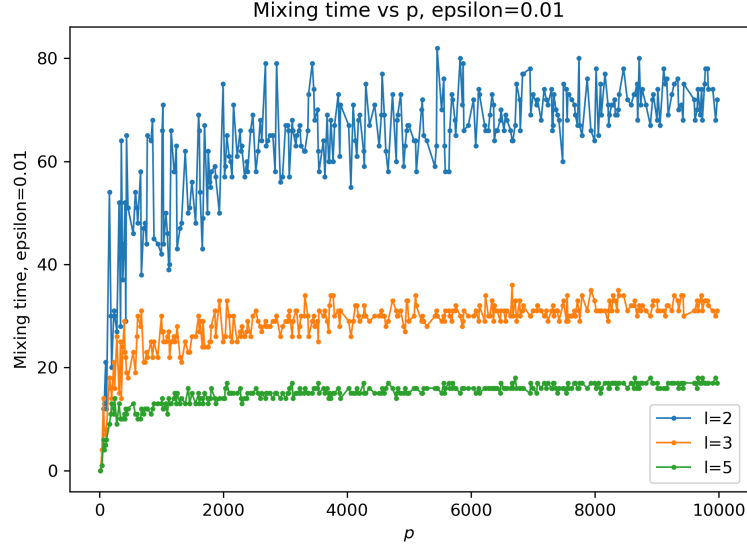


Figure 4.13: Mixing times $t_{\text{mix}}(0.01)$ plotted against increasing p .

Figures 4.11, 4.12 and 4.13 appear to demonstrate something approximating a logarithmic relationship between mixing times and p , for all values ℓ and ε . Indeed, this observation is borne out by a result that mixing times should have asymptotic class $O(\log(p))$. Our empirical results demonstrate both the adherence to this asymptotic relation, as well as the large fluctuations that can occur locally, between nearby values p .

Since supersingular ℓ -isogeny graphs have size N_p growing linearly with p , one would expect in general that mixing times increase according to p . But, due to their strong connectivity properties, we see that this growth is slow—indeed logarithmically slow—which is to say that the number of steps required for random walks to approach uniformity does not increase quickly as p increases.

We also see that mixing times are lower for larger ℓ , which coheres again with our observation that a larger ℓ implies a greater number of edges over the same number of vertices. Indeed, this should produce a better connected graph, and therefore random walks should more quickly diffuse into uniformity.

Since we have that mixing times are given asymptotically by $O(\log(p))$, and that $p \approx N_p$, we should expect that by plotting the same mixing times against $\log(N_p)$ instead of p , we would witness a nearly linear relationship. This does appear to nearly follow, observing Figures 4.14 4.15 and 4.16, excepting the noise we already observed.

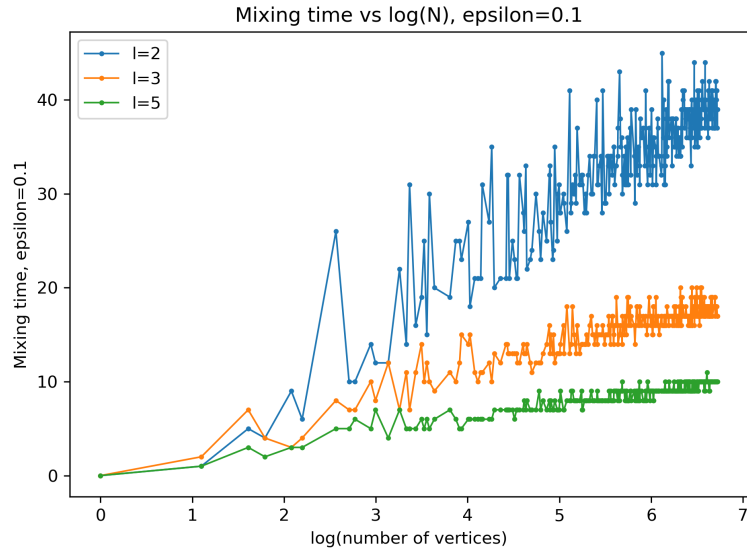


Figure 4.14: Mixing times $t_{\text{mix}}(0.1)$ plotted against increasing $\log(N_p)$.

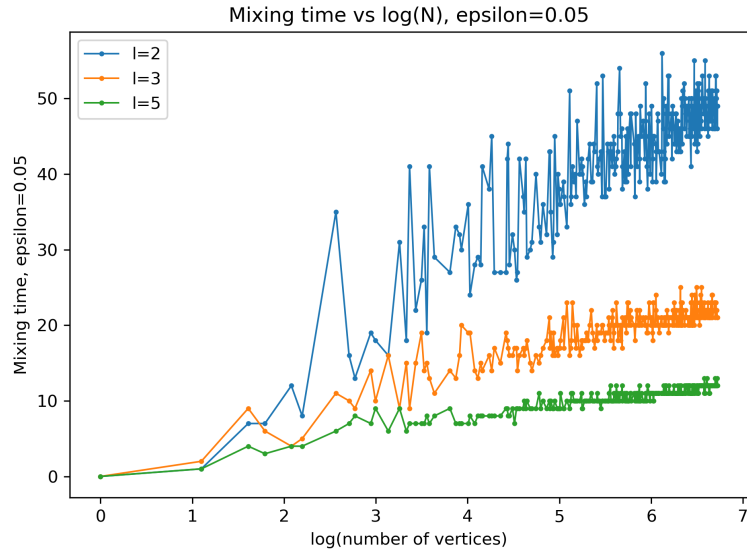


Figure 4.15: Mixing times $t_{\text{mix}}(0.05)$ plotted against increasing $\log(N_p)$.

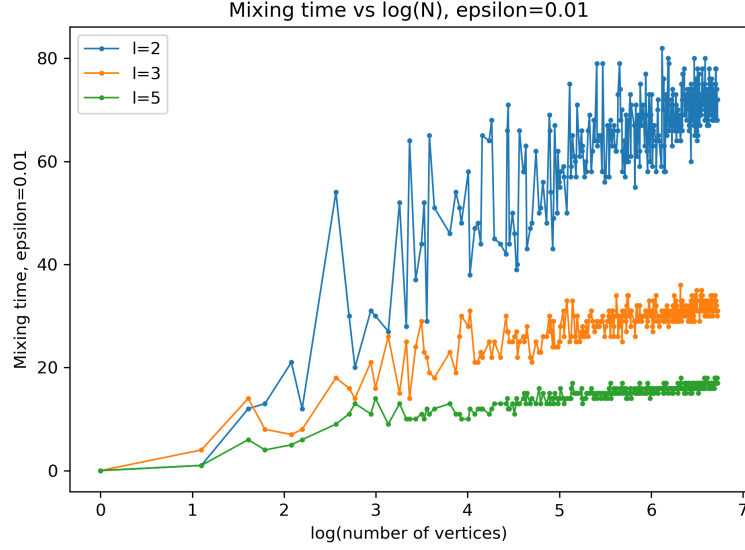


Figure 4.16: Mixing times $t_{\text{mix}}(0.01)$ plotted against increasing $\log(N_p)$.

Finally, in Figure 4.17, we observe the total variation distance $d_{\text{TV}}(P_n, U)$ plotted against increasing step count n , in the cases of the curves $p = 4789$ and $p = 9973$ with $l \in \{2, 3, 5\}$. Here there appears to be a very clear, linear decline in $d_{\text{TV}}(P_n, U)$ in all cases, up to a point at which the curve appears to reach a ‘floor’ and the curve plateaus. We see in these examples that higher ℓ appears to correspond with a faster convergence of ℓ toward 0.

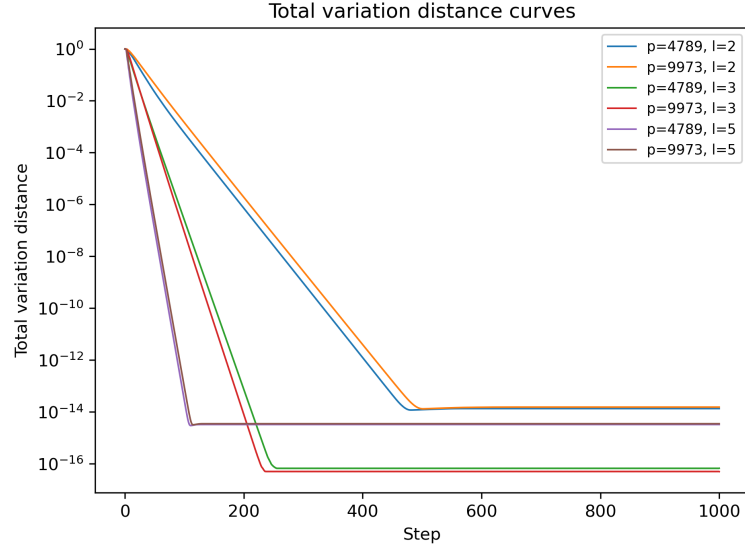


Figure 4.17: Values of the total variation distance $d_{\text{TV}}(P_n, U)$ plotted against increasing step count n , in the cases of the curves $p = 4789$ and $p = 9973$ with $l \in \{2, 3, 5\}$.

Following the definition of the total variation distance, we see that this plot demonstrates rapid convergence of random walks toward the uniform distribution. It can be suggested that the point at which each curve appears to plateau indicates the number of steps after which random walks are adequately mixed, with further steps providing negligible mixing benefit.

4.5 Random walks and empirical total variation distance

In the preceding section, we used numerical computation of the probability distribution P_n to ascertain mixing times and total variation distances according to standard analytical tools. However, it is also

often informative to see results such as these illustrated by concrete examples. To this end, we have produced a sample of random walks, from which we derive a Monte-Carlo illustration of mixing over supersingular isogeny graphs.

We simulated for each $l \in \{2, 3, 5\}$ and $p \in \{4789, 9973\}$ a sample of 5000 random walks for lengths

$$n \in \{1, 2, 3, 5, 10, 20, 30, 50\}.$$

From this we obtained for each above n an empirical probability distribution

$$\hat{P}_n(i) = \frac{\text{\#sampled walks ending at vertex } i}{\text{\#sampled walks}},$$

using which we were able to compute an *empirical* total variation distance $d_{\text{TV}}(\hat{P}_n, U)$. The curve of this value against increasing walk lengths is shown in Figure 4.18.

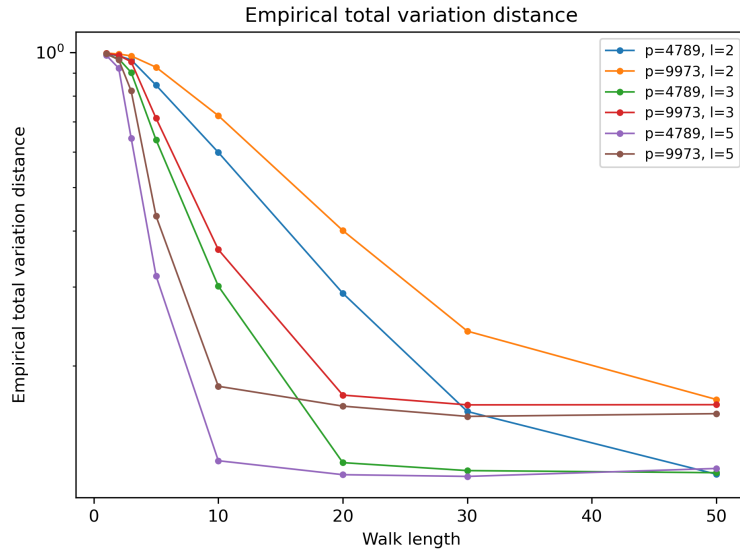


Figure 4.18: Values of the empirical total variation distance $d_{\text{TV}}(\hat{P}_n, U)$, plotted against number of steps n .

In Figure 4.18, we see that the empirical total variation distance declines monotonically in all the sampled cases p and l . While not matching Figure 4.17 precisely, this is to be expected due to noise in finite samples. While we will prefer the analytical values obtained in 4.17 for accuracy, these empirical plots provide qualitative corroboration of the general phenomena at hand.

Additionally, for each of the selected representative graphs, we computed 5000 random walks of length 50 steps, all from a common initial vertex for each graph, in order to obtain an empirical estimate for the frequency of each vertex as an endpoint. If it is true that after 50 steps the random walks have adequately mixed, each histogram should show something resembling a uniform distribution with height $1/N_p$ at each point. Excepting some noise, this appears to be what we observe in Figures 4.19 to 4.22, where in each case the expected values for a uniform distribution have been marked by a horizontal dotted line.

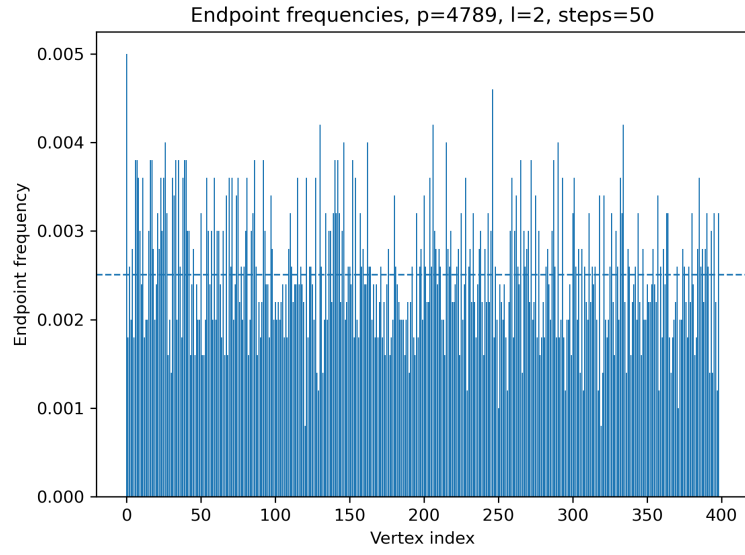


Figure 4.19: Histogram of endpoints for random walks on $G_\ell^s(\mathbb{F}_{p^2})$ after 50 steps, for $p = 4789$ and $\ell = 2$.

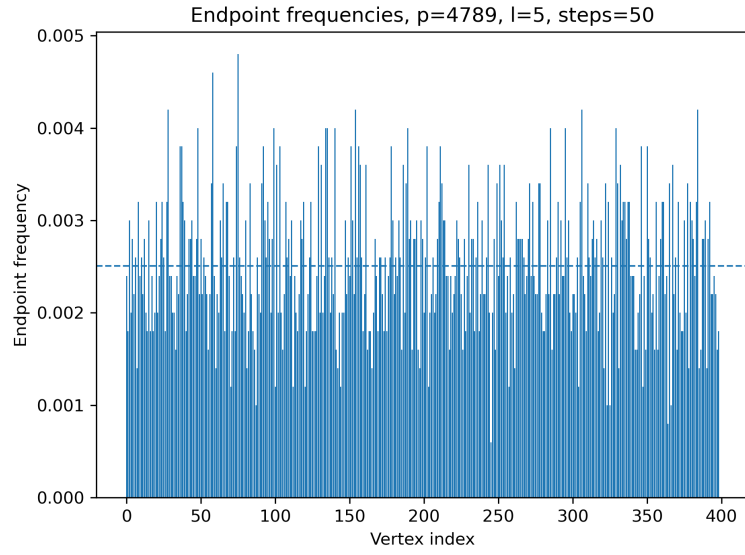


Figure 4.20: Histogram of endpoints for random walks on $G_\ell^s(\mathbb{F}_{p^2})$ after 50 steps, for $p = 4789$ and $\ell = 5$.

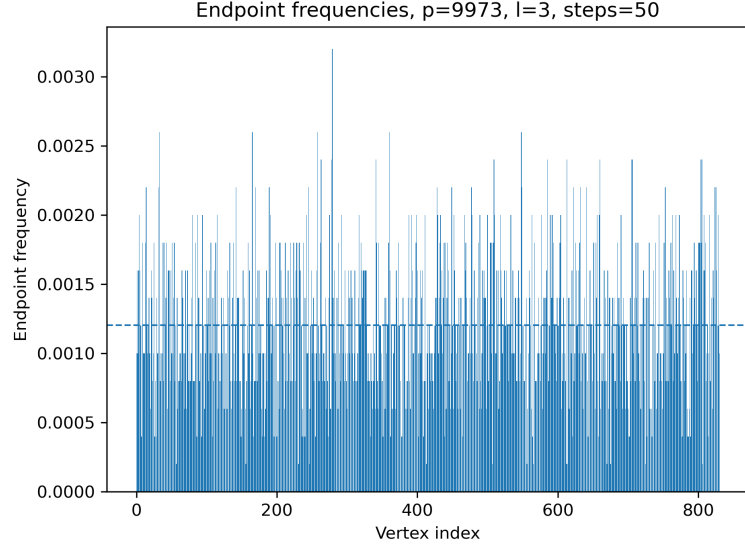


Figure 4.21: Histogram of endpoints for random walks on $G_\ell^s(\mathbb{F}_{p^2})$ after 50 steps, for $p = 9973$ and $\ell = 3$.

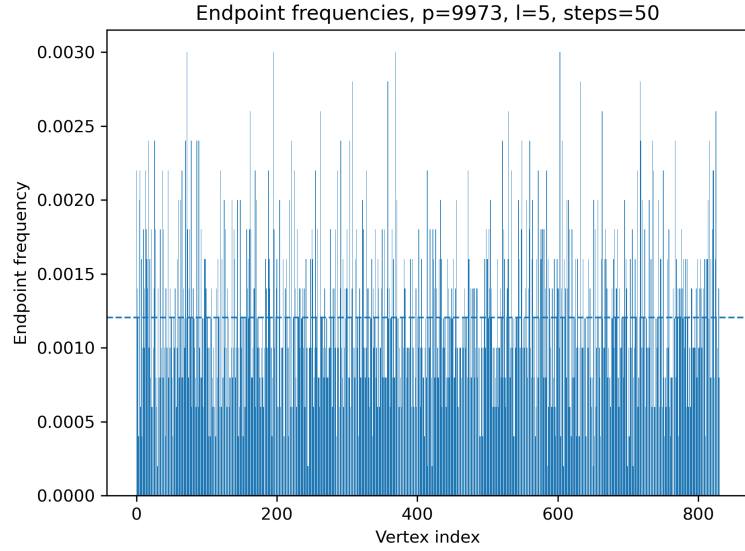


Figure 4.22: Histogram of endpoints for random walks on $G_\ell^s(\mathbb{F}_{p^2})$ after 50 steps, for $p = 9973$ and $\ell = 5$.

4.6 Comparing the exact and empirical total variation distances

Finally, we will like to compare the total variation distances obtained using the exact and empirical means. The same qualitative phenomena were observed for graphs with both $p = 4789$ and $p = 9973$, so we will attach only the latter for brevity.

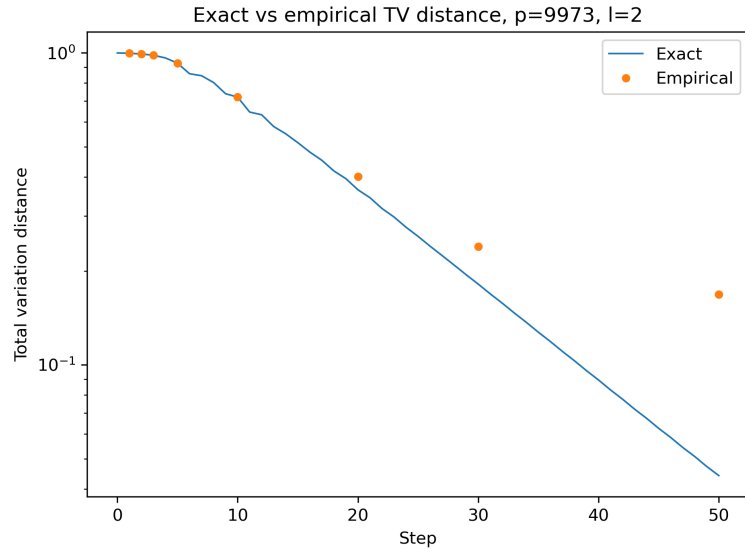


Figure 4.23: Comparison of empirical and exact total variation distances for $p = 9973$ and $\ell = 2$.

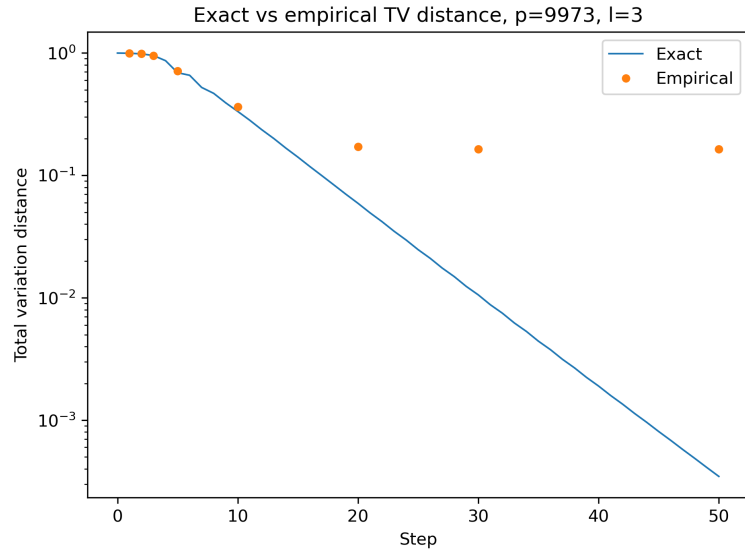


Figure 4.24: Comparison of empirical and exact total variation distances for $p = 9973$ and $\ell = 3$.

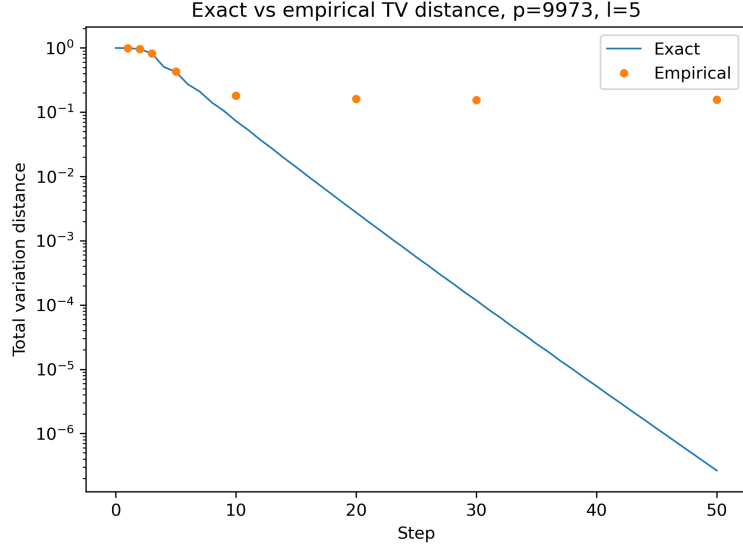


Figure 4.25: Comparison of empirical and exact total variation distances for $p = 9973$ and $\ell = 5$.

Here, in Figures 4.23, 4.24 and 4.25, we observe an interesting discrepancy. While for $\ell = 2$ it appears that the exact and empirical total variation distances are fairly aligned, as ℓ increases to values 3 and 5, the empirical values slowly peel upwards, failing to follow the downward pattern followed by the exact curve. I am not yet certain why this is the case, except that I hypothesise this is related to having inadequately large samples. Indeed, we have sampled 5000 random walks, but $N_{9973} = 831$. Therefore, we expect on average that each vertex has been sampled only $5000/831 \approx 6.02$ times. This is, despite our efforts, still quite a small sample. For the case that $p = 4789$, we still have $N_{4789} = 399$, and therefore expect each vertex to have been visited an average of $5000/399 \approx 12.53$ times. This is an improvement over the case that $p = 9973$, but still potentially inadequate for strong estimates.

However, even if this discrepancy can be attributed to inadequate sample sizes, it fails to explain the consistent trend observed across our selected samples that the empirical total variation distance peels *upwards* from the exact total variation distance curve, rather than having symmetric error about the expected value. I speculate that an explanation for this effect will require mathematical examination of the total variation distance.

Chapter 5

Conclusion

Summarise findings, suggest further possible questions.

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